

Aerospace nozzles are critical components in rocket and jet propulsion systems because they convert high-pressure, high-temperature gases into high-velocity exhaust to produce thrust. The geometry of a nozzle directly affects efficiency, thrust output, and performance at different altitudes. Traditional **converging-diverging (bell) nozzles** are commonly used and are designed to perform best at **specific atmospheric conditions**, but their efficiency decreases when operating outside those conditions. In contrast, **aerospike nozzles** are advanced designs that allow exhaust gases to expand externally along a **central spike**, enabling better adaptation to changing **atmospheric pressure** and improving overall efficiency. This feature, known as **altitude compensation**, allows aerospike nozzles to maintain more consistent performance across a range of altitudes.

Designing and analyzing aerospace nozzle geometries using ANSYS enables [comparison](#) of traditional bell nozzle and an aerospike nozzle to evaluate how shape affects airflow, pressure, velocity, and thrust. Using [CFD simulations](#), the goal is to understand nozzle efficiency and show how [geometry](#) impacts propulsion performance and can improve efficiency in aerospace applications.

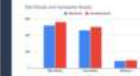


Max Velocity	520 m/s	560 m/s
Exit Mach	1.9	2.1
Efficiency	88%	94%
Mass Flow Rate	0.85 kg/s	0.90 kg/s

Figure 9 shows the bell nozzle's thermal contour, with rapid expansion and a quick temperature drop, indicating faster energy loss and lower efficiency.



Figure 11 compares maximum velocity, showing that the aerospike nozzle achieves **higher** velocity than the bell nozzle, indicating more efficient **acceleration** and better **overall performance**.



Wang, J. T., Sato, A. & Yuan, H. J. (2008) A new linear approach to linear Algebraic Nizkor. *Journal of Probability and Power*, 7(1), 10–18. <https://doi.org/10.1016/j.jpp.2007.12.001>